5. Evidence of Disparate Privacy Across Clinical Populations

5.1 Underrepresented Populations and Rare Clinical Conditions

Underrepresentation, clinical rarity, and record atypicality are related but should not be treated as interchangeable. An underrepresented group contributes a small number or proportion of observations to the model's training data. A rare condition is a diagnosis, phenotype, or clinical state with low prevalence in the relevant population or dataset. An atypical record, by contrast, occupies a low-density or unusual region of the feature distribution and may belong to either a minority or majority group. A rare condition may generate relatively typical records within its clinical subgroup, whereas a common condition can contain highly atypical individual presentations. Evidence concerning these three properties should therefore be assessed separately.

Knolle et al. provide the most direct clinical patient-linked evidence linking limited representation to extreme membership vulnerability [3]. Across seven medical datasets, they estimated record-level MIA AUC using 200 repeatedly trained target models and defined the extreme-vulnerability tail as the highest 1% of record-level AUC values. They then compared the subgroup composition of this tail with the composition expected from the complete training data. Significant differences were observed for several stratifications, including disease status, self-reported race, sex, insurance status, and imaging protocol. In MIMIC-IV-ED, records from Black patients and patients with Medicaid insurance occurred 31% and 126% more frequently than expected in the tail, respectively. Across all subgroup comparisons, large positive Pearson residuals occurred mainly among groups contributing less than 20% of the records, and group size was weakly to moderately inversely correlated with tail enrichment [3]. These results support an association between limited representation and disproportionate extreme record-level vulnerability.

The evidence concerning rare diseases and clinical states is more heterogeneous. In EMBED, benign findings classified as BI-RADS-2 and findings suspicious of malignancy classified as BI-RADS-4 were overrepresented in the extreme tail by 60% and 1,179%, respectively; almost entirely fatty and extremely dense breasts were also disproportionately represented [3]. However, the model predicted breast density rather than tumour status, and the reported racial differences in this dataset may themselves be confounded by density. In MIMIC-IV-ED, cancer records were 18% more frequent than expected in the tail, but the cancer comparison was not statistically significant after correction. Statistical significance could not be evaluated for disease-label comparisons in CheXpert and MIMIC-CXR because patients could have multiple, non-mutually-exclusive diagnoses. Conversely, marked disease-related composition differences were reported in datasets such as FairVision and Fitzpatrick 17k. Collectively, these findings indicate dataset-specific associations, but do not establish that rare diagnoses universally produce higher privacy vulnerability.

Nonclinical and health-administrative studies provide supporting, rather than directly transferable, evidence. Kulynych et al. defined subgroup vulnerability as normalized membership advantage within a population group and introduced repeated-measures tests for comparing it reliably [7]. On Adult, smaller racial groups such as Asian-Pac-Islander and "Other" exhibited significant disparities under neural-network models. On Texas-50K, which was derived from hospital-discharge records, nearly all subgroup pairs differed under the 32-neuron network. Nevertheless, standard logistic regression did not show significant disparity, demonstrating that representation alone was insufficient and that model family and subgroup-specific generalization mattered. Chang and Shokri further manipulated group proportions in synthetic data and found that reducing the size of an unprivileged group increased its fairness-induced privacy cost [18]. Their Law, Bank, and COMPAS experiments likewise showed that privacy effects varied with the protected attribute, model capacity, and fairness constraint. These are record-level benchmark results and should not be presented as evidence about rare clinical conditions.

A clinical-language-model preprint also examines disease rarity and group-level membership, but does not supply a diagnostic-model multi-record patient effect [50]. Several alternative explanations constrain causal interpretation. Smaller groups may have higher loss, poorer predictive performance, greater feature variability, or different label distributions. Disease, demographics, insurance, and acquisition protocol may also overlap, making marginal subgroup comparisons difficult to interpret. Moreover, Knolle et al.'s disparity analysis used records, although patients may contribute different

numbers of correlated records. Let T denote membership in the selected extreme-vulnerability tail. Composition tests compare $P(G = g \mid T)$ with the group's overall representation $P(G = g)$. With the same record as the unit, enrichment also implies an increased probability $P(T \mid G = g)$ for that particular tail, but it does not show that every group member has elevated continuous vulnerability or that patient-level vulnerability is higher when record multiplicity differs. The relative top-1% definition also creates an extreme tail in every model, irrespective of its absolute security. Small subgroup counts, repeated comparisons, and threshold choice introduce additional uncertainty, although Knolle et al. applied row-wise Bonferroni correction and Kulynych et al. used ANOVA with multiplicity-adjusted post-hoc tests [3,7].

5.2 Demographic and Intersecting Subgroups

Evidence of demographic disparity in membership vulnerability must distinguish among three analytically different comparisons. A marginal analysis compares groups defined by one attribute, such as race, sex, or insurance status. A label-conditional analysis compares an attribute within levels of the clinical outcome or prediction label. Genuine intersectional analysis instead evaluates combinations of two or more social or demographic attributes, such as race × sex × insurance status. This distinction matters because a protected attribute combined with a target label does not constitute a demographic intersection. Moreover, race, ethnicity, and insurance status should be interpreted as socially and institutionally situated variables rather than biological patient characteristics.

Knolle et al. found record-level top-1% tail enrichment in several clinical demographic strata [3]. In MIMIC-IV-ED, Black-patient and Medicaid records were 31% and 126% more prevalent in that tail than their dataset shares; effects varied by attribute and dataset. These marginal record-composition tests do not estimate patient-level identification at a fixed FPR or intersections among demographic attributes. Repeated records per patient can also change representation. Tier A thus supports heterogeneity, not a universal demographic ranking.

Health-administrative and nonclinical studies provide supporting, but less directly transferable, evidence. Kulynych et al. found statistically significant racial differences in normalized membership-inference advantage for some neural-network configurations on the Adult and Texas-50K hospital-discharge datasets [7]. However, disparities varied with model family and were often absent for logistic regression, indicating that demographic vulnerability was not an invariant property of a group. Zhong et al. similarly evaluated gender and race separately, conditioning each attribute on the binary target label [19]. On Adult, aggregate MIA accuracy was 54.1%, whereas accuracy reached 69.8% for one race–label subgroup; the largest corresponding between-group gap was 12.5 percentage points. Smaller groups were often more vulnerable, but exceptions occurred, and group size alone did not explain the results. The Hospital dataset supplied a health-related administrative setting, yet its unit remained a discharge record rather than a patient, and neither study evaluated intersections among multiple demographic attributes.

Research on algorithmic fairness further shows that subgroup conclusions depend on the attack and operating point. Chang and Shokri demonstrated that fairness interventions could change loss-based membership vulnerability differently across protected-attribute–label groups [18]. Their subgroup thresholds, however, assumed substantial attacker knowledge, and their experiments were not clinical. Purkait et al. subsequently evaluated subgroup-tailored likelihood-ratio attacks at very low FPRs [21]. In Bank Marketing, subgroup-restricted sampling produced a TPR of 13.6% at an observed 0% FPR for one subgroup, compared with 12.2% using population sampling; under adversarial debiasing, the corresponding values were 9.1% and 3.5%. These results show that population-level attacks can miss subgroup-specific leakage. Nevertheless, parameters were optimized separately for each subgroup, and the subgroup method underperformed LiRA when a single global configuration was imposed. The study therefore informs audit design rather than establishing clinical or intersectional prevalence.

Patient-level demographic intersections remain underpowered and sparsely studied. Herlihy et al.'s diabetes utility study illustrates the small-cell problem—8 of 32 race–sex–age cells had fewer than ten observations—but does not estimate privacy [20]. Future intersectional audits should prespecify cells, account for patient clustering and missingness, test site and disease confounding, and report intervals and multiplicity-aware comparisons under Table 4.1.

5.3 Clinical Settings, Institutions, and Acquisition Protocols

Clinical context can enter a privacy audit at three related but distinct levels. The first is the institution or site, including the hospital, clinical centre, geographical region, or dataset source. The second is acquisition, encompassing scanner or device vendor, imaging view, scan settings, sampling frequency, and ECG lead configuration. The third is clinical workflow, such as inpatient versus outpatient care, emergency versus scheduled assessment, screening versus diagnostic imaging, and the period in which data were collected. These factors may change what the target model learns, the baseline difficulty of individual records, and the correspondence between target and reference data. Consequently, an attack may separate members from non-members because training induced record-specific behavior, because the two groups originate from different clinical environments, or through a combination of both. Attack separability therefore should not be interpreted as pure membership leakage without examining site and acquisition confounding.

Knolle et al. found protocol-associated variation in the record-level top-1% tail across chest-radiograph views and mammography projections [3]. Their chi-squared comparisons with row-wise Bonferroni correction establish associations, not protocol causation or patient-level identification rates. View and projection may correlate with mobility, disease, workflow and protocol frequency; controls must separate these explanations.

The distinction between membership and institutional origin becomes particularly important when members and non-members come from different datasets. Wang et al. evaluated self-supervised ECG encoders using subject-level aggregation and reported AUC, TPR at a 1% FPR, and attacker advantage [22]. Leakage varied substantially across training objectives and cohorts and was especially pronounced in some small, institution-specific datasets. However, non-members were sampled from ECG datasets disjoint from the pretraining dataset. The authors acknowledge that the resulting signal may combine subject participation with acquisition hardware, voltage scaling, sensor noise, resampling, or other dataset-specific properties; they accordingly interpret the results as cross-dataset participation leakage rather than pure within-dataset subject identification. Humphries et al. demonstrate the same identification problem more directly [23]. When a target model was trained using heart-disease records from Cleveland, Hungary, Switzerland, or VA Long Beach and records from the other hospitals were treated as non-members, membership advantage was higher than under an IID construction formed by shuffling the hospital datasets. The attack may therefore be exploiting a property shared by records from the training hospital. Such site recognition can itself reveal sensitive institutional or cohort participation, but it does not answer the narrower question of whether a particular patient participated among other patients from the same institution.

Data-partition artifacts introduce a related threat to validity. Bai et al. evaluated an attack involving synthetic brain MRI images generated by StyleGAN3 and added a member-only negative control in which the classifier distinguished two disjoint subsets of training patients, despite both subsets having identical membership status [24]. The control achieved higher AUC than the primary member–non-member experiment in both the small setting (0.8421 versus 0.8216) and the large setting (0.9768 versus 0.9655), with bootstrap confidence intervals for both differences excluding zero. This result shows that high discrimination may arise from learnable partition structure rather than model leakage. Its direct generalizability is limited because it concerns a generative model and synthetic MRI matching, but the negative-control principle applies broadly. Shi et al. further show that mismatch between target and shadow distributions has no uniform effect: cutout, jitter, Gaussian noise, label shift, and attribute shift could either increase or decrease attack performance, and the same shift could have opposite effects for different MIAs [25]. Thus, lower success under site shift does not necessarily demonstrate stronger privacy, while higher success does not by itself establish patient memorization.

Within-site member/non-member contrasts, site-matched reference models and member-only or permutation controls can distinguish patient participation from institutional recognition [22–25]. Disease, acquisition, demographics and patient record count should be considered in these comparisons. Cross-dataset participation remains a distinct outcome. Table 4.1 gives the common patient-level reporting protocol; Section 5.4 asks how exposure varies within sites and groups.

5.4 Within-Group Variation and Extreme-Risk Patients

Group-level disparity and within-group heterogeneity answer different questions. Between-group disparity concerns differences in average vulnerability or in the composition of a high-vulnerability tail across populations, whereas within-group variation concerns how widely vulnerability differs among patients assigned to the same population. An extreme-vulnerability tail denotes the upper part of an estimated vulnerability distribution rather than an intrinsic or permanent patient characteristic. Its membership is conditional on the attack, target model, membership hypothesis, unit of analysis, reference distribution, training repetitions, and prespecified threshold or percentile. Consequently, a subgroup with modest average vulnerability may still contain highly exposed patients, while a subgroup with elevated average vulnerability may include many patients whose estimated vulnerability remains close to chance.

The strongest direct clinical patient-linked evidence comes from Knolle et al. [3]. Across seven clinical datasets, the authors trained 200 target models on varying training subsets and estimated a record-level MIA AUC for each record. Patient-level exposure was then summarized using the maximum record-level MIA AUC among the records contributed by that patient. Empirical survival functions revealed a small tail of patients with near-perfect distinguishability even when aggregate attack AUC remained close to random guessing, demonstrating marked heterogeneity that system-level averages failed to represent. This finding nevertheless requires three qualifications. First, taking a maximum creates more opportunities for an extreme value as record count increases; patient-level tail membership may therefore be partly influenced by record multiplicity, although this effect was not separately quantified in the study. Second, the subgroup analysis of the highest 1% was conducted at the record level and therefore does not directly estimate patient-level tail probability. Third, the study did not formally decompose total variability into within-group and between-group components. It therefore supports substantial patient-level heterogeneity, but not the stronger claim that within-group variation exceeds between-group disparity.

Group identity alone is consequently insufficient for identifying an extreme-risk patient. Let $\mathcal{T}$ denote the prespecified extreme-vulnerability tail and V the vulnerability estimate for the declared unit of analysis. Tail composition measures

$$P(G = g \mid V \in \mathcal{T}),$$

whereas within-group tail prevalence measures

$$P(V \in \mathcal{T} \mid G = g).$$

These quantities are related through group prevalence but are not interchangeable. A large group may constitute a substantial portion of the tail even when each member has a relatively low tail probability; conversely, a small group may have high within-group tail prevalence while contributing few patients to the tail. Moreover, absence of group overrepresentation does not imply absence of extremely vulnerable individuals. Kulynych et al. show that naïvely applying membership attacks can overestimate subgroup disparity and provide a statistical framework for testing such differences [7]. This concern is especially consequential when subgroup counts are small.

General benchmarks support heterogeneous example-level vulnerability but do not determine its clinical magnitude: Song and Mittal show fine-grained differences [15], Carlini et al. show why low-FPR results matter [2], Watson et al. separate difficulty from membership [13], and Jälkö et al. caution against globally pooled per-target thresholds and finite-population bias [14].

Clinical persistence is unresolved. On nonclinical benchmarks, binary LiRA high-vulnerability sets at very low FPR reproduced less well across seeds than continuous likelihood-ratio rankings [26]. The clinical extreme tail therefore warrants intervals and repeated, comparable audits; neither a single top-1% list nor a group mean identifies permanently exposed patients.

5.5 Consistency, Conflicting Findings, and Limits of Generalization

Consistency has at least three meanings in this literature. Within-study consistency asks whether a finding persists across datasets, models, training repetitions, or attacks under one protocol. Between-

study consistency requires independent investigations to recover comparable patterns, whereas transportability asks whether a result remains valid in a new institution, population, or deployment setting. These levels should not be conflated. The multi-dataset result discussed above is one coordinated study, not independent external replication. Moreover, membership-inference evaluations may implement different experiments and uncertainty sources [6]; comparison is meaningful only when the protected secret, membership hypothesis, access assumptions, and estimand are sufficiently aligned.

Across evidence tiers, sample-specific leakage, low-FPR exposure and subgroup variation recur [2,3,7,15]. Knolle et al. provide the direct diagnostic-model patient-linked result, but the nonclinical studies are methodological corroboration, not independent estimates of the clinical tail's size (Table 3.2).

Agreement weakens regarding the direction and identity of disparity. Underrepresentation is not a universal predictor: Section 5.1 shows that direction and significance varied by dataset and attribute. Kulynych et al. found meaningful disparities but warned that naïve application of existing attacks may overestimate them [7]. Similarly, Chang and Shokri [18], Zhong et al. [19], and Purkait et al. [21] do not establish a fixed ranking of protected groups. Fairness-aware training may increase, decrease, or redistribute membership vulnerability depending on the algorithm, subgroup, model, and attack [18,21]. Such variation is better interpreted as context dependence than as evidence that disparity is absent. Conversely, non-significance does not establish equal privacy, particularly when groups are small, tail events are rare, or multiplicity correction reduces power.

Many apparent conflicts instead reflect different estimands. Record-level vulnerability is not equivalent to patient-level membership vulnerability; as distinguished in Section 5.4, top-tail composition differs from the probability that a member of a particular group enters that tail; and aggregate ROC-AUC answers a different question from TPR at a prespecified low FPR [2,3,6,7]. Uncalibrated scores may also reflect intrinsic sample difficulty rather than membership-specific behavior [13], while concatenating scores across targets need not control the FPR for each sample [14]. Likewise, within-site auditing distinguishes members among comparable patients, whereas cross-dataset auditing may exploit hospital, device, or preprocessing signals. This distinction affects the subject-level ECG results of Wang et al. [22] and the hospital-source effects examined by Humphries et al. [23]. Bai et al. showed that split artifacts can rival a nominal MIA signal [24], while Shi et al. found that distribution shift can either strengthen or weaken attacks [25]. Findings based on these different contrasts are not necessarily contradictory.

Generalization remains restricted by few independent patient-level audits, limited institutions and model families, sparse intersectional evidence, and inconsistent membership games. Cross-dataset ECG non-members confound subject participation with cohort origin [22]; low-FPR binary sets can also be unstable across runs [26]. Table 3.2 marks which studies directly estimate clinical patients and which supply methods or analogies.

The protected event, analysis unit, comparator population and operating point differ too much for quantitative pooling. The evidence supports heterogeneous, sometimes disparate exposure, while a universal ranking of clinical populations or stable identification of the same patients remains unestablished.